

4.5 Manipulating Power Series

Recall: There are functions (such as $f(x) = e^{-x^2}$) which have no antiderivative of elementary functions.

Can we use the knowledge we have learned so far in Chapter 4 to find power series representations for these kinds of functions?

Motivation: We proved in the previous section that the function $f(x) = \sin(x)$ was equal to its Maclaurin series for all x . That is,

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Use this information to find a power series for $g(x) = \cos(x)$.

We can do this by differentiating the Maclaurin series of $\sin(x)$ in particular:

$$\begin{aligned} \cos(x) &= \frac{d}{dx}(\sin(x)) = \frac{d}{dx} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{2n}}{(2n)!} \right). \end{aligned}$$

By the theorem from the start of this section we know that this series converges for all x , and cosine is equal to its Maclaurin series for all x .

The following table is a list of important Maclaurin series that one should know or be able to quickly derive.

Function and Maclaurin Series	Interval of Convergence
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$	$(-1, 1)$
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$	$(-\infty, \infty)$
$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	$(-\infty, \infty)$
$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$(-\infty, \infty)$
$\tan^{-1}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	$[-1, 1]$
$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$	$(-1, 1]$

There are several things we can do to manipulate given power series representations to obtain new power series representations.

We can Manipulate by:

- 1) Seeing functions as sums of geometric series.
- 2) Taking derivatives and antiderivatives of series & functions.
- 3) Multiplying by polynomials.
- 4) Substituting polynomials (this may change the IOC & ROC).

Example 1: Find a power series representation for $f(x) = \frac{x-2}{(1-x)(1+x)}$

We can see that $f(x) = \frac{x-2}{(1-x)(1+x)} = \frac{(x-2)}{1-x^2}$. This resembles the sum of a geometric series with

$a = (x-2)$ & $r = x^2$. Therefore $f(x) = \frac{(x-2)}{1-x^2} = \sum_{n=0}^{\infty} (x-2)(x^2)^n$ for $|x^2| < 1 \rightarrow -1 < x < 1$.

Example 2: Consider the series $\sum_{n=1}^{\infty} nx^n$.

a) Determine a function modeled by the series $\sum_{n=1}^{\infty} nx^n$ and state its radius of convergence.

We know that $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for x values in $(-1,1)$ and therefore $R=1$.

We take the derivative of both sides and obtain $\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}$ which also has $R=1$.

We then see that if both sides are multiplied by x we obtain $x \cdot \sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2} \rightarrow \sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$

Therefore $\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$ with $R=1$.

b) Determine the sum of the series $\sum_{n=1}^{\infty} n\left(\frac{1}{2}\right)^n$.

We see that this is the series from part (a) being evaluated at $x = \frac{1}{2}$. Since this value is inside the interval of convergence for this series we can say that $\sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n = \frac{1/2}{\left(1 - \frac{1}{2}\right)^2} = \frac{1/2}{1/4} = 2$

Example 3: Find the Maclaurin series for $f(x) = x \cos(x)$

By what we have shown in the previous example we know that

$$f(x) = x \cos(x) = x \left(\sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{2n}}{(2n)!} \right) \right) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{2n+1}}{(2n)!} \right) \text{ which is the Maclaurin series for } f(x) = x \cos(x).$$

Note: Like Example 3, the Taylor series of products of functions with known Taylor series can be found by performing products or quotients of these power series. E.g., $\tan(x) = \sin(x)/\cos(x)$.

Example 4: Find a series that sums to equal $\frac{\pi}{6}$.

We notice that $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$ and since $\frac{1}{\sqrt{3}}$ is within the interval $[-1, 1]$ we can say that

$$\sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{\sqrt{3}}\right)^{2n+1}}{2n+1} = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

- Why would this problem need to be solved differently if we wanted a series that summed to be $\frac{\pi}{3}$?

We notice that $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$. However $\sqrt{3}$ is *not* within the interval $[-1, 1]$ and so we cannot create a series using the inverse tangent that would sum to this value. Finding such a series is possible to do using a geometric series.

$$\text{Notice that } \frac{\pi}{3} = \frac{\pi/6}{1 - (1/2)} = \sum_{n=0}^{\infty} \frac{\pi}{6} \left(\frac{1}{2}\right)^n.$$

- Find a series that sums to $\frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right)$.

We know that $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = \tan^{-1}(x)$ for x values in $[-1, 1]$. Therefore if we multiply both sides by x we obtain $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+1} = x \tan^{-1}(x)$. Since $x = \frac{1}{2}$ is inside the IOC we can conclude that

$$\sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{2}\right)^{2n+2}}{2n+1} = \frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right)$$

Recall: The function $f(x) = e^{-x^2}$ has no antiderivative of elementary functions. However, this function is fundamental to understanding statistics and probability.

Example 5: Represent $\int e^{-x^2} dx$ exactly as an infinite series.

By what has been shown in this class we know that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all x . This means that

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}. \text{ By the theorem from this section we know that } \int e^{-x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}.$$

We know also by the same theorem that this series converges for all x .

Example 6: Use the result of example 5 to approximate the integral $\int_0^1 e^{-x^2} dx$ using a Maclaurin series.

In example 5 we found $\int e^{-x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}$. Therefore, with a definite integral, we know that:

$$\int_0^1 e^{-x^2} dx = \left[\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!} \right]_0^1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)n!} - 0 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)n!}. \text{ Any degree of accuracy can be calculated for this integral!}$$

- Determine how close the 2nd partial sum of the series is to the exact value of $\int_0^1 e^{-x^2} dx$.

Since the series found above is an alternating series we can use the theorem about the sum of an alternating series from 3.6. Recall that for an alternating series with sum S it is the case that

$$|s_n - S| \leq a_{n+1}$$

We note here that $s_1 = \frac{1}{1 \cdot 1} + \frac{-1}{3 \cdot 1} = \frac{2}{3}$ and that $a_2 = \frac{1}{5 \cdot 2} = \frac{1}{10}$

Therefore we can say that $\left| \frac{2}{3} - S \right| \leq \frac{1}{10}$, that is, the value of $\frac{2}{3}$ is at most $\frac{1}{10}$ away from the actual sum S .